Phase-erased arrows of time and entropy production under periodic driving: a bound under known time symmetry, a no-go without it, and the locality of the pairwise estimator

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1 October 2026
Preprint
DOI: 10.5281/zenodo.23049617

Abstract

We ask when forward records with the drive phase erased can bound entropy production in periodically driven finite-state Markov jump processes. Known time symmetry and a reversal-covariant observation give a finite-window relative-entropy bound with a mirror-state boundary allowance; the corresponding rate bound has no boundary term. The allowance vanishes exactly when all frozen generators share the periodic state's constant stationary law, a weaker condition than detailed balance. Without time symmetry, a two-state construction fixes the four-sample arrow at any prescribed value in an explicit interval while the window's expected entropy production tends to zero. This excludes a universal quantitative lower bound from that arrow alone, not qualitative or model-specific inference. Exact probability tables and numerical checks accompany the construction. The two-state witness's pairwise waiting-time estimator is identically zero at every period. A separate locality result covers graphs with hidden edges under explicit uniformity hypotheses; it does not prove Maier and collaborators' finite-period pairwise-estimator conjecture. The ingredients of the time-symmetric bound are classical; no priority claim is made.

1 Introduction

When does an arrow visible after forgetting drive phase bound total entropy production? The distinction between finite-window boundary terms and long-time rates is essential, as is the distinction between reversal of a recorded trajectory and execution of the reversed drive protocol. The following conditions let a forward-only, phase-erased arrow bound total entropy production; the conditions are sufficient (Theorem T1), not necessary for every forward-only inference:

(1) the protocol is known to be time-symmetric up to a phase shift, $Q(\theta) = Q(c - \theta)$;
(2) the observation map is reversal-covariant, $O \circ R = R' \circ O$;
(3) finite windows are charged the boundary allowance $\bar{B}$.

Without known time symmetry no universal certificate $L\bar{\sigma} \geq F(\mathrm{KL}_L^{\mathrm{obs}})$ from this arrow alone exists (Theorem T2). A universal *quantitative* lower bound from the arrow alone then needs information beyond the arrow; the routes available here are known time symmetry, reversed-protocol data, or phase-resolved occupations together with transition rates. Qualitative and model-specific forward-only inferences are not excluded: in the two-state class of T2 a positive arrow still implies $\bar{\sigma} > 0$ (Remark 4.4).

Classical baseline. The ingredients of Theorem T1 are a forward/reversed-protocol likelihood identity and information contraction. Related classical results are Crooks's fluctuation theorem [1], the Hamiltonian phase-space dissipation identity of Kawai, Parrondo and Van den Broeck [2], and the coarse-graining bounds of Gomez-Marin, Parrondo and Van den Broeck [3]. The finite-state path identity needed here is derived explicitly below.

Verification material. The supplement [6] supplies exact probability tables, numerical checks, source code and a history note identifying invalid earlier runs. Section 7 separates these checks from the analytic arguments. An earlier version used the three-sample comparator in a weaker limiting argument; the theorem below uses the reversal-covariant four-sample arrow and an exact intermediate-value construction.

2 Setting

2.1 Periodically driven jump processes, phase erasure, observation maps

Definition 2.1 (Driven process, window entropy production)**.** Let S be a finite set of states, each even under time reversal, and let $Q(\theta)$, $\theta \in \mathbb{R}/\mathcal{T}\mathbb{Z}$, be a $\mathcal{T}$-periodic family of generators ($Q_{ij}(\theta) \geq 0$ for $i \neq j$, rows summing to zero), with bounded, right-continuous, piecewise continuously differentiable rates and finitely many switching phases per period, irreducible, with $Q_{ij}(\theta) > 0$ if and only if $Q_{ji}(\theta) > 0$, and with a unique periodic state $p(\theta)$ solving the forward equation $\dot{p} = pQ$ almost everywhere with $p(\theta) > 0$ componentwise. Derivatives at switches are interpreted one-sidedly; finite entropy comparisons assume integrability of the displayed rate-ratio terms. For a window $[\theta_0, \theta_0 + L]$ let P_{θ_0} be the law of the path $(x_t)_{0 \leq t \leq L}$ of the process started in $p(\theta_0)$ and driven by $Q(\theta_0 + t)$. Crooks's total entropy production of the window is

$$\Sigma_{\mathrm{tot}}(L; \theta_0) = \ln p_{\theta_0}(x_0) - \ln p_{\theta_0 + L}(x_L) + \sum_{\text{jumps } x_- \to x_+ \text{ at } t} \ln \frac{Q_{x_- x_+}(\theta_0 + t)}{Q_{x_+ x_-}(\theta_0 + t)}, \tag{1}$$

the system term plus the medium term. Writing $\sigma(\theta) = \sum_{i<j} \big(p_i Q_{ij} - p_j Q_{ji}\big) \ln \frac{p_i Q_{ij}}{p_j Q_{ji}}(\theta) \geq 0$ for the total entropy production rate in the periodic state, one has $\mathbb{E}\, \Sigma_{\mathrm{tot}}(L; \theta_0) = \int_0^L \sigma(\theta_0 + t)\, dt$, and the period-averaged rate is $\bar{\sigma} = \mathcal{T}^{-1} \int_0^{\mathcal{T}} \sigma(\theta)\, d\theta$. Since the system entropy is periodic, $\bar{\sigma}$ is also the period-averaged medium entropy production rate.

Definition 2.2 (Phase erasure, covariant observation, observed arrow)**.** The starting phase θ_0 is uniform on $[0, \mathcal{T})$ and independent of the noise; the phase-erased path law is $\bar{P}_L = \mathcal{T}^{-1} \int_0^{\mathcal{T}} P_{\theta_0}\, d\theta_0$. Let R be path reversal on the window, $(Rx)_t = x_{L-t}$. An observation map O from paths to records is *phase-free* if it does not use θ_0 and *reversal-covariant* if there is an involution R' of records with $O \circ R = R' \circ O$. Examples: the state path itself (R' = path reversal); the states sampled on a grid symmetric under $t \mapsto L - t$ (R' = order reversal); memoryless state-dependent emissions with time-even marks; transition records with a declared mark involution. Detector memory, first-event selection and dead time can break covariance; they require a separate check of the observation map. The observed arrow and the full arrow are

$$\mathrm{KL}_L^{\mathrm{obs}} := D\big(\bar{P}_L^{\mathrm{obs}} \,\big\|\, R'_* \bar{P}_L^{\mathrm{obs}}\big), \qquad \mathrm{KL}_L^{\mathrm{full}} := D\big(\bar{P}_L \,\big\|\, R_* \bar{P}_L\big), \qquad \bar{P}_L^{\mathrm{obs}} = O_* \bar{P}_L, \tag{2}$$

where $D(P\|Q) = \int \ln \frac{dP}{dQ}\, dP$ is relative entropy. Both are computed from the forward experiment alone: no reversed protocol is run.

Definition 2.3 (Time symmetry, boundary allowance)**.** The protocol is *time-symmetric* if there is a c with $Q(\theta) = Q(c - \theta)$ almost everywhere. Equalities between protocols are understood in this sense; values at individual switching times do not change the path law. For such c the boundary allowance is

$$\bar{B} := \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} D\big(p(\psi) \,\big\|\, p(c - \psi)\big)\, d\psi \;\geq 0, \tag{3}$$

which is finite when $p > 0$. It measures how far the periodic state is from its mirror image.

2.2 The two-state model

Definition 2.4 (Two-state family)**.** Take $S = \{0, 1\}$, $Q_{01}(t) = u(t)$, $Q_{10}(t) = 1 - u(t)$, with u $\mathcal{T}$-periodic and $0 < u < 1$; the total rate is 1. Let $p(t)$ be the periodic probability of state 1; it solves $\dot{p} = u - p$, so $p(t) = \int_0^\infty e^{-r} u(t - r)\, dr$. For the trigonometric drive

$$u(t) = m + e_1 \cos \omega t + e_2 \cos(2\omega t + \varphi), \qquad \omega = 2\pi/\mathcal{T}, \tag{4}$$

the periodic state is in closed form,

$$p(t) = m + e_1 A_1 \cos(\omega t - \alpha_1) + e_2 A_2 \cos(2\omega t + \varphi - \alpha_2), \qquad A_k = \big(1 + (k\omega)^2\big)^{-1/2}, \quad \alpha_k = \arctan(k\omega). \tag{5}$$

The entropy production rate is

$$\sigma(t) = \big(u(t) - p(t)\big) \ln \frac{u(t)\,(1 - p(t))}{(1 - u(t))\,p(t)} \;\geq 0, \tag{6}$$

and $\mathcal{T}\bar{\sigma} = \int_0^{\mathcal{T}} \sigma(t)\,dt$ is the entropy production per period. For $e_1 e_2 \neq 0$ the drive (4) is time-symmetric if and only if $\varphi \equiv 0 \pmod{\pi}$, and then $c = 0$. With $c = 0$ the allowance (3) reads $\bar{B} = \mathcal{T}^{-1} \int_0^{\mathcal{T}} \big[p \ln \frac{p}{p^-} + (1-p) \ln \frac{1-p}{1-p^-}\big](t)\,dt$ with $p^-(t) = p(-t)$.

Definition 2.5 (Phase-erased three-sample comparator)**.** The propagator of the two-state process is $\Pr(X_{t_1} = 1 \mid X_{t_0} = x_0) = p(t_1) + \big(x_0 - p(t_0)\big)e^{-(t_1 - t_0)}$. For lags $a, b > 0$ let $\bar{P}_{a,b}(x_0, x_1, x_2)$ be the phase-erased law of $\big(X_{\theta_0}, X_{\theta_0 + a}, X_{\theta_0 + a + b}\big)$, that is

$$\bar{P}_{a,b}(x_0, x_1, x_2) = \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} \pi_{x_0}(t)\, G(x_1 \mid x_0; t, t + a)\, G(x_2 \mid x_1; t + a, t + a + b)\,dt, \tag{7}$$

with $\pi_1 = p$, $\pi_0 = 1 - p$ and G the propagator. The *phase-erased three-sample comparator* is

$$\mathrm{KL}_3(a, b) := \sum_{x \in \{0,1\}^3} \bar{P}_{a,b}(x_0, x_1, x_2) \ln \frac{\bar{P}_{a,b}(x_0, x_1, x_2)}{\bar{P}_{b,a}(x_2, x_1, x_0)}. \tag{8}$$

Its quasi-static counterpart at phase lags α, β is

$$c_*(\alpha, \beta) := \sum_x P^{\mathrm{qs}}_{\alpha,\beta}(x_0, x_1, x_2) \ln \frac{P^{\mathrm{qs}}_{\alpha,\beta}(x_0, x_1, x_2)}{P^{\mathrm{qs}}_{\beta,\alpha}(x_2, x_1, x_0)}, \qquad P^{\mathrm{qs}}_{\alpha,\beta}(x) = \frac{1}{2\pi} \int_0^{2\pi} \prod_{k=0}^{2} \mathrm{Bern}\big(u(\theta + \tau_k), x_k\big)\,d\theta, \tag{9}$$

where $(\tau_0, \tau_1, \tau_2) = (0, \alpha, \alpha + \beta)$, $u(\theta) = m + e_1 \cos\theta + e_2 \cos(2\theta + \varphi)$ in phase units, and $\mathrm{Bern}(q, 1) = q$, $\mathrm{Bern}(q, 0) = 1 - q$. We also write $c_3^* := c_*$.

Definition 2.6 (Covariant four-sample arrow)**.** For lags $a, b > 0$ and $L = a + b$ let O_4 record the states on the reflection-symmetric grid $\{0, a, b, L\}$ (in increasing order), with R' the order reversal of records. The *covariant four-sample arrow* is its observed arrow in the sense of Definition 2.2,

$$\mathrm{KL}_4(a, b) := \mathrm{KL}_L^{\mathrm{obs}} = D\big(O_{4*}\bar{P}_L \,\big\|\, R'_* O_{4*} \bar{P}_L\big),$$

and its quasi-static counterpart at phase lags α, β, $c_4^*(\alpha, \beta)$, is defined as in (9) with the four phases $(0, \alpha, \beta, \alpha + \beta)$ and the reversed record in the denominator.

Lemma 2.7 (KL_3 is dominated by the covariant arrow)**.** *Let $L = a + b$ and let O_3 sample the state at times 0, a and $a + b$. Then*

$$\mathrm{KL}_3(a, b) = D\big(O_{3*}\bar{P}_L \,\|\, O_{3*} R_* \bar{P}_L\big) \;\leq\; \mathrm{KL}_4(a, b) \;\leq\; \mathrm{KL}_L^{\mathrm{full}}.$$

For $a \neq b$ the map O_3 is not reversal-covariant, so KL_3 is a forward-only comparator (both of its laws are forward phase-erased laws, at lag orders (a, b) and (b, a)), not an observed arrow in the sense of Definition 2.2; it is the $(0, a, L)$-marginal comparison of the covariant arrow KL_4.

Proof. The reversed path sampled at 0, a, $a + b$ is $(O_3 \circ R)(x) = (x_L, x_{L-a}, x_0) = (x_{a+b}, x_b, x_0)$, which uses the unrecorded state x_b. Its law under $\bar{P}_L$, evaluated at (x_0, x_1, x_2), is $\bar{P}_{b,a}(x_2, x_1, x_0)$, the denominator in (8). For the grid $\{0, a, b, a + b\}$, reversal maps (x_0, x_a, x_b, x_L) to (x_L, x_b, x_a, x_0), an order reversal since $L - a = b$; the marginal of $R'_* O_{4*} \bar{P}_L$ on the coordinates $(0, a, L)$ is again $\bar{P}_{b,a}(x_2, x_1, x_0)$, and the marginal of $O_{4*}\bar{P}_L$ is $\bar{P}_{a,b}$. Data processing for this marginal gives the first inequality, and for O_4 the second. □

Remark 2.8 (Strict inequality at the witness)**.** At the witness of Section 4.2 the quasi-static four-sample law, $(m, e_1, e_2, \varphi) = (9/20, 1/4, 1/5, \pi/3)$ on the phases $(0, \pi/3, 2\pi/3, \pi)$, has sixteen positive rational cells with denominator 640000 (Appendix A). The records 0000 and 0010 both project to the triple 000. Record 0000 has forward and reversed probabilities 50681/640000 and 50681/640000; record 0010 has 42799/640000 and 45799/640000. So the likelihood ratio takes the two values 1 and 42799/45799 on one fibre. By the equality case of data processing, $c_4^* > c_3^*$ strictly; in fact every one of the eight fibres is non-constant. Two computational routes reproduce this exact witness. The values are

$$c_3^* = 0.000738887716627336479919968717667\ldots, \qquad c_4^* = 0.001534419994182709441167521655 04\ldots,$$

agreeing to 45 digits on the two routes. The independently evaluated 30-digit quadrature result differs by relative error $1.5 \cdot 10^{-28}$.

Remark 2.9 (Moment structure of the quasi-static comparator)**.** The eight probabilities $P^{\mathrm{qs}}_{\alpha,\beta}(x)$ are determined by the moments $\mathbb{E}u_k$, $\mathbb{E}u_k u_l$, $\mathbb{E}u_0 u_1 u_2$ of $u_k = u(\theta + \tau_k)$ under uniform θ. The single moments are all m; the pair moments are values of the autocorrelation $\mathbb{E}u(\theta)u(\theta+\tau) = m^2 + \frac{1}{2}e_1^2\cos\tau + \frac{1}{2}e_2^2\cos 2\tau$, even in τ, at the same three lags $\alpha, \beta, \alpha+\beta$ for the forward and the reversed law. The two laws can differ only through the triple moment, $\mathbb{E}\,u(\theta)u(\theta+\alpha)u(\theta+\alpha+\beta)$ against $\mathbb{E}\,u(\theta)u(\theta+\beta)u(\theta+\alpha+\beta)$, and within it only through the (1,1,2)-harmonic term proportional to $e_1^2 e_2$, the only one carrying the biphase φ. At $\varphi \equiv 0 \pmod{\pi}$ the two agree (time symmetry); at the lags $(\alpha, \beta) = (\pi/2, \pi)$ they agree identically in φ. This (1,2)-harmonic cancellation gives $c_* = 0$ at those lags even for $\varphi = \pi/3$, providing a zero control for the numerical checks.

3 Theorem T1: known time symmetry survives phase erasure

Theorem 3.1 (T1)**.** *Let Q be time-symmetric, $Q(\theta) = Q(c-\theta)$, with periodic state $p > 0$, and let O be phase-free and reversal-covariant. Then for every window length $L > 0$,*

$$\mathrm{KL}_L^{\mathrm{obs}} \;\le\; L\,\bar{\sigma} + \bar{B}. \tag{10}$$

Proof. Step 1 (the reversed experiment is a forward experiment). The reversed protocol of the window $[\theta_0, \theta_0 + L]$ is $\tilde{Q}(t) = Q(\theta_0 + L - t)$, $0 \le t \le L$. By symmetry, $Q(\theta_0 + L - t) = Q(c - \theta_0 - L + t)$, so the process under the reversed protocol, started in the periodic state at its initial phase, is the forward process started at phase $c - \theta_0 - L$; its law is $P_{c-\theta_0-L}$.

Step 2 (Radon–Nikodym derivative). Against the reversed paths of that process, the log-likelihood ratio of a forward path is $\ln p_{\theta_0}(x_0) - \ln p_{c-\theta_0-L}(x_L)$ plus the medium term of (1), the rates $Q_{ij}(\theta) > 0 \Leftrightarrow Q_{ji}(\theta) > 0$ making it finite. Adding and subtracting $\ln p_{\theta_0+L}(x_L)$ and taking the P_{θ_0}-expectation,

$$D\big(P_{\theta_0} \,\big\|\, R_* P_{c-\theta_0-L}\big) = \mathbb{E}\,\Sigma_{\mathrm{tot}}(L;\theta_0) + D\big(p(\theta_0+L) \,\big\|\, p(c-\theta_0-L)\big), \tag{11}$$

because x_L has law $p(\theta_0 + L)$ under P_{θ_0}.

Step 3 (phase average). Average (11) over uniform θ_0. On the right, $\mathbb{E}_{\theta_0}\mathbb{E}\,\Sigma_{\mathrm{tot}}(L;\theta_0) = \mathbb{E}_{\theta_0}\int_0^L \sigma(\theta_0+t)\,dt = L\bar{\sigma}$ (the system term averages to zero by periodicity), and the substitution $\psi = \theta_0 + L$ turns the second term into $\bar{B}$. On the left, the map $\theta_0 \mapsto c - \theta_0 - L$ preserves the uniform law on the circle, so $\mathcal{T}^{-1}\int R_* P_{c-\theta_0-L}\,d\theta_0 = R_*\bar{P}_L$, and joint convexity of relative entropy gives $\mathrm{KL}_L^{\mathrm{full}} = D(\bar{P}_L \| R_*\bar{P}_L) \le L\bar{\sigma} + \bar{B}$.

Step 4 (data processing). $D(O_*\bar{P}_L \| O_* R_* \bar{P}_L) \le D(\bar{P}_L \| R_* \bar{P}_L)$, and by covariance $O_* R_* \bar{P}_L = R'_* O_* \bar{P}_L = R'_* \bar{P}_L^{\mathrm{obs}}$. Since O is phase-free it commutes with the phase mixture, so the left side is $\mathrm{KL}_L^{\mathrm{obs}}$. □

Corollary 3.2 (Phase loss and observation loss under known symmetry)**.** *Under the hypotheses of Theorem 3.1, assume the relative entropies below are finite. On phase–path space define*

$$\mathbb{P}(d\theta, dx) = \frac{d\theta}{\mathcal{T}} P_\theta(dx), \qquad \mathbb{Q}(d\theta, dx) = \frac{d\theta}{\mathcal{T}} (R_* P_{c-\theta-L})(dx).$$

Let P_X, Q_X be their path marginals and P_Y, Q_Y the marginals of $Y = O(X)$. Then

$$L\bar{\sigma} + \bar{B} = \mathrm{KL}_L^{\mathrm{obs}} + I_{\mathrm{phase}} + I_{\mathrm{obs}}, \tag{12}$$

where the two nonnegative terms are

$$I_{\text{phase}} = \int D\big(\mathbb{P}(d\theta \mid x) \,\big\|\, \mathbb{Q}(d\theta \mid x)\big)\, P_X(dx),$$
$$I_{\text{obs}} = \int D\big(P_X(dx \mid y) \,\big\|\, Q_X(dx \mid y)\big)\, P_Y(dy).$$

These quantify the losses from first erasing phase and then reducing the path to the recorded observable, in that order.

Proof. The phase marginals of $\mathbb{P}$ and $\mathbb{Q}$ are the same uniform law. Steps 2–3 of the proof of Theorem 3.1 therefore give $D(\mathbb{P}\|\mathbb{Q}) = L\bar{\sigma} + \bar{B}$. The relative-entropy chain rule, first conditioned on X and then on $Y = O(X)$, gives

$$D(\mathbb{P}\|\mathbb{Q}) = D(P_X\|Q_X) + I_{\text{phase}} = D(P_Y\|Q_Y) + I_{\text{obs}} + I_{\text{phase}}.$$

The phase shift in $\mathbb{Q}$ preserves the uniform law, so $Q_X = R_*\bar{P}_L$; reversal covariance identifies $D(P_Y\|Q_Y) = \mathrm{KL}_L^{\text{obs}}$. □

This is a chain-rule consequence in the known-symmetry setting, not a new universal entropy identity. In particular the reversed joint law uses $c - \theta - L$, not the unshifted phase. The finite-window allowance $\bar{B}$ remains part of the left side. The two losses depend on this order of coarse-graining and do not assert two independent physical mechanisms. The decomposition does not extend the bound to the unknown-symmetry counterexample.

Corollary 3.3 (rate form and score form)**.** *Under the hypotheses of Theorem 3.1:*

(a) (rate form) $\bar{\sigma} \;\geq\; \limsup_{L\to\infty} \dfrac{\mathrm{KL}_L^{\text{obs}}}{L}$*, with no boundary term;*

(b) (score form) for any record score S *with values in* $[-1,1]$ *that is odd under* R'*,* $S \circ R' = -S$*, with phase-erased mean* $\mu_L = \mathbb{E}_{\bar{P}_L^{\text{obs}}} S$*,*

$$L\,\bar{\sigma} \;\geq\; 2\mu_L^2 - \bar{B}. \tag{13}$$

Proof. (a) Divide (10) by L; $\bar{B}$ is finite and independent of L. (b) $\mathbb{E}_{\bar{P}^{\text{obs}}} S - \mathbb{E}_{R'_*\bar{P}^{\text{obs}}} S = \mu_L - \mathbb{E}_{\bar{P}^{\text{obs}}}(S \circ R') = 2\mu_L$, while $|\mathbb{E}_P S - \mathbb{E}_Q S| \leq 2\,\mathrm{TV}(P,Q)\sup|S|$; hence $\mathrm{TV} \geq |\mu_L|$ and Pinsker's inequality $D \geq 2\,\mathrm{TV}^2$ gives $2\mu_L^2 \leq \mathrm{KL}_L^{\text{obs}} \leq L\bar{\sigma} + \bar{B}$. □

The rate form needs no model-dependent allowance. Finite windows must be charged $\bar{B}$, which is model-dependent and bounded by population ratios when $p \geq p_{\min} > 0$; short windows can be dominated by it (Section 7 shows a window where it is comparable to $L\bar{\sigma}$).

Proposition 3.4 (When the allowance vanishes)**.** *Let* $Q(\psi) = Q(c-\psi)$ *and* $p > 0$*. Then* $\bar{B} = 0$ *if and only if the periodic state is constant,* $p(\psi) \equiv \pi$*, with* $\pi Q(\psi) = 0$ *for every* ψ*; that is, if and only if* p *is a common stationary law of every frozen generator.*

Proof. $\bar{B} = 0$ forces $D(p(\psi)\|p(c-\psi)) = 0$ for almost every ψ, hence $p(\psi) = p(c-\psi)$ for all ψ by continuity. Differentiating almost everywhere, $\dot{p}(\psi) = -\dot{p}(c-\psi) = -p(c-\psi)Q(c-\psi) = -p(\psi)Q(\psi) = -\dot{p}(\psi)$, so $p \equiv \pi$ with $\pi Q(\psi) = 0$ almost everywhere, and hence everywhere by right continuity of Q. Conversely a constant p equals its mirror image and $\bar{B} = 0$. □

Remark 3.5 (stationarity is weaker than detailed balance)**.** Common *detailed balance*, $\pi_i Q_{ij}(\psi) = \pi_j Q_{ji}(\psi)$, and preparation in π guarantee same-experiment path reversal for a palindromic schedule: the reversed path weights agree by detailed balance and the equal integrated escape rates. This is strictly stronger than Proposition 3.4. For example, a three-state cycle with constant clockwise rate k_+ and counterclockwise rate k_-, $k_+ \neq k_-$, is a (constant, hence time-symmetric) protocol whose uniform stationary law is common to every frozen generator, so $\bar{B} = 0$, while detailed balance fails and $\bar{\sigma} > 0$; Theorem 3.1 then reads $\mathrm{KL}_L^{\text{obs}} \leq L\bar{\sigma}$.

4 Theorem T2: no-go without time symmetry

Theorem 4.1 (T2)**.** *Let $\delta = (2-\sqrt{3})/8$ and let $\mathcal{M}_\delta$ be the class of two-state periodically driven models of Definition 2.4 whose rate $u = Q_{01}$ and periodic population p lie in $[\delta, 9/10]$ (so every rate and population lies in $[\delta, 1-\delta]$), observed through the covariant four-sample record of Definition 2.6 with lags $(\mathcal{T}/6, \mathcal{T}/3)$ and $L = \mathcal{T}/2$. Let $F\colon (0,\infty) \to \mathbb{R}$ be any function, with no continuity or monotonicity assumed. If $F(k_0) > 0$ for some $k_0 \in (0, c_4^*)$, where $c_4^* = c_4^*(\pi/3, 2\pi/3)$ at the witness drive, then some model in $\mathcal{M}_\delta$ without time symmetry has $\mathrm{KL}_L^{\mathrm{obs}} = k_0$ exactly and $L\bar{\sigma} < F(k_0)$. Hence no certificate $L\,\bar{\sigma} \geq F(\mathrm{KL}_L^{\mathrm{obs}})$ holds on $\mathcal{M}_\delta$ without knowledge of time symmetry for any F with $F(k) > 0$ for all $k > 0$. A universal lower bound therefore needs information beyond this arrow; the routes available here are known time symmetry (Theorem 3.1), reversed-protocol data [5], or phase-resolved occupations and transition rates [4, Eq. 19]. The same holds for the three-sample comparator KL_3 with $c_3^* = c_*$ in place of c_4^*.*

Remark 4.2 (Why an exact fixed-arrow construction is needed)**.** A sequence with positive limiting arrow and vanishing dissipation excludes lower-bound functions bounded away from zero near that limiting arrow. It does not exclude an arbitrary positive function with no such regularity. The construction below keeps the arrow exactly fixed, eliminating that restriction. For $F(k) = k$, the finite-period example in Table 1 already violates the proposed bound.

4.1 Exact fixed-arrow construction

Proof of Theorem 4.1. Consider the drives

$$u_\varphi(\theta) = \tfrac{1}{4}(1+\cos\theta) + \tfrac{1}{5}\big(1+\cos(2\theta+\varphi)\big), \qquad 0 \leq \varphi \leq \pi/3,$$

that is $(m, e_1, e_2) = (9/20, 1/4, 1/5)$: the witness at $\varphi = \pi/3$, time-symmetric at $\varphi = 0$. Let $K_\mathcal{T}(\varphi)$ be the covariant four-sample arrow $\mathrm{KL}_4(\mathcal{T}/6, \mathcal{T}/3)$ of this drive.

Step 1. *Rates.* $u_\varphi \leq \frac{1}{2} + \frac{2}{5} = \frac{9}{10}$. If the circular distance from θ to π is at least $\pi/6$, then $\cos\theta \geq -\cos(\pi/6)$ and the first term is at least $\delta = (2-\sqrt{3})/8$. Otherwise $\theta = \pi + \varepsilon$ with $|\varepsilon| < \pi/6$, so $2\theta + \varphi \equiv 2\varepsilon + \varphi \in (-\pi/3, 2\pi/3)$, $\cos(2\theta+\varphi) > -\frac{1}{2}$ and the second term exceeds $\frac{1}{10} > \delta$. Hence $\delta \leq u_\varphi \leq \frac{9}{10}$ for all θ and all $\varphi \in [0, \pi/3]$.

Step 2. *Populations.* $p(t) = \int_0^\infty e^{-r} u(t-r)\, dr$ is an average of values of u, so $\delta \leq p \leq \frac{9}{10}$.

Step 3. *Lag.* $|u - p| \leq \int_0^\infty e^{-r} |u(t) - u(t-r)|\, dr \leq \sup|\dot{u}| \int_0^\infty r e^{-r}\, dr \leq (\frac{1}{4} + \frac{2}{5})\omega = \frac{13}{20}\,\omega$.

Step 4. *Dissipation.* By (6), $\sigma = (u-p)(\mathrm{logit}\, u - \mathrm{logit}\, p)$, and on $[\delta, \frac{9}{10}]$ the derivative of logit is $1/(x(1-x)) \leq 10/\delta$. Hence $\sigma \leq (10/\delta)(\frac{13}{20})^2\omega^2$ and, with $L = \mathcal{T}/2$ and $\omega = 2\pi/\mathcal{T}$,

$$L\,\bar{\sigma} \;\leq\; \frac{169\,\pi^2}{20\,\delta\,\mathcal{T}} \qquad \text{uniformly in } \varphi \in [0, \pi/3]. \tag{14}$$

Step 5. *Continuity.* For fixed $\mathcal{T}$ the sixteen record probabilities are phase averages of products of the periodic population and transition probabilities of an irreducible two-state chain with rates $\geq \delta$; they are positive and continuous in φ, so $K_\mathcal{T}$ is continuous on $[0, \pi/3]$.

Step 6. *Limits.* As $\mathcal{T} \to \infty$ at fixed φ, $e^{-\mathcal{T}/6} \to 0$ and $p \to u$ uniformly (Step 3), so the record law converges to the positive quasi-static law and $K_\mathcal{T}(\varphi)$ converges to the relative entropy between that law and its reversal. At $\varphi = 0$ the drive is even and the grid is reflection-symmetric, so the quasi-static law is reversal-invariant and $K_\mathcal{T}(0) \to 0$; at $\varphi = \pi/3$, $K_\mathcal{T}(\pi/3) \to c_4^* > 0$.

Step 7. *Intermediate values.* Fix $k_0 \in (0, c_4^*)$. For all large $\mathcal{T}$, $K_\mathcal{T}(0) < k_0 < K_\mathcal{T}(\pi/3)$, and the intermediate value theorem gives $\varphi_\mathcal{T} \in (0, \pi/3)$ with $K_\mathcal{T}(\varphi_\mathcal{T}) = k_0$ exactly. That model is not time-symmetric ($\varphi_\mathcal{T} \notin \pi\mathbb{Z}$, $e_1 e_2 \neq 0$), and by (14) its $L\bar{\sigma} \to 0$, so $L\bar{\sigma} < F(k_0)$ once $\mathcal{T}$ is large. This proves Theorem 4.1; the comparator case runs identically on the $(0, \pi/3, \pi)$ marginals, whose quasi-static comparison is c_* at $\varphi = \pi/3$ and 0 at $\varphi = 0$.

□

The argument is an existence proof: it gives no explicit $\varphi_\mathcal{T}$ and no calibration prescription, it covers the declared two-state class only, and prescribed arrows $k_0 \geq c_4^*$ are not covered by this family. The constant in (14) is loose (at $\mathcal{T} = 40960$ it is 0.0608 against the computed $L\bar{\sigma} = 0.000264$), but it tends to zero, which is all the argument needs. As an illustration only (one route, not part of the proof), at $\mathcal{T} = 40960$ the exact finite-period law gives

$K_{\mathcal{T}}(0) = 1.06 \cdot 10^{-25}$ and $K_{\mathcal{T}}(\pi/3) = 0.0015344197795$, so the bracket $K_{\mathcal{T}}(0) < c_*/2 < K_{\mathcal{T}}(\pi/3)$ already holds there.

Remark 4.3 (Scope of T1)**.** The hypotheses of Theorem 3.1 (known time symmetry, a phase-free reversal-covariant observation) are sufficient for the bound. Their failure does not forbid every model-specific forward-only inference; T2 concerns a universal lower bound from this arrow alone.

Remark 4.4 (qualitative implications survive T2)**.** Theorem 4.1 excludes universal *quantitative* lower bounds positive somewhere in $(0, c_4^*)$; it does not remove qualitative or model-specific implications. In $\mathcal{M}_\delta$ itself, $\bar{\sigma} = 0$ forces $\sigma \equiv 0$, because the integrand (6) is ≥ 0; then $u \equiv p$, since logit is strictly increasing; then $\dot{p} = u - p = 0$, so the drive is constant. The process is then a stationary time-homogeneous two-state chain, which is reversible, so $\mathrm{KL}_L^{\mathrm{obs}} = 0$. Hence in $\mathcal{M}_\delta$ a positive observed arrow does imply $\bar{\sigma} > 0$. What fails is a universal function with $F(k_0) > 0$ at a prescribed $k_0 \in (0, c_4^*)$ bounding $L\bar{\sigma}$ from below. Whether a qualitative implication of this kind holds beyond the two-state class is not addressed here.

4.2 A finite-period witness

The witness is the drive

$$u(t) = 0.45 + 0.25 \cos \omega t + 0.2 \cos(2\omega t + \pi/3), \tag{15}$$

that is $(m, e_1, e_2, \varphi) = (0.45, 0.25, 0.2, \pi/3)$, with lags $(a, b) = (\mathcal{T}/6, \mathcal{T}/3)$, so $L = \mathcal{T}/2$ and $(\alpha, \beta) = (\pi/3, 2\pi/3)$, and $\mathcal{T} \in \{40, 160, 640, 2560, 10240, 40960\}$. A phase grid estimates the rate extrema as approximately 0.0997 and 0.8742; the analytic positive lower bound used in the theorem is δ, not the grid estimate. Table 1 shows the $1/\mathcal{T}$ decay of entropy production per period and convergence of the comparator to its positive quasi-static value. Full-precision outputs are supplied in `supplement/results/EP1B_RUN2.json`.

Table 1: The asymmetric two-state witness. The observation window has length $L = \mathcal{T}/2$. Values are rounded from the supplied numerical record.

$\mathcal{T}$	EP per period	$\mathcal{T}$·EP	KL_3, the comparator	$L\bar{\sigma}$ over the window
40	0.498	19.93	$6.43 \cdot 10^{-4}$	0.249
160	0.134	21.47	$7.32 \cdot 10^{-4}$	0.0671
640	0.0337	21.60	$7.385 \cdot 10^{-4}$	0.0169
2560	0.00844	21.608	$7.3886 \cdot 10^{-4}$	0.00422
10240	0.00211	21.610	$7.38886 \cdot 10^{-4}$	0.00106
40960	0.000528	21.6105	$7.38888 \cdot 10^{-4}$	**0.000264**

At $\mathcal{T} = 40960$, the comparator is approximately 0.000738887612, whereas $L\bar{\sigma}$ is approximately 0.000263800024. Thus $\mathrm{KL}_4 \geq \mathrm{KL}_3 > L\bar{\sigma}$. The positive limiting arrow describes the drive's time asymmetry, even as dissipation per period tends to zero. Every frozen generator of this two-state model satisfies detailed balance; the period-scale arrow does not establish broken detailed balance at a frozen phase.

5 Local waiting-time statistics in the slow-driving limit

Maier, Degünther, van der Meer and Seifert [4] distinguish three estimators. Their reversed-protocol waiting-time bound is proved (journal Eq. 16); their forward-only phase-averaged estimator is conjectured to be a bound (Eqs. 17–18); and their occupation-and-rate bound is proved without reversing the protocol (Eq. 19). These are Eqs. 13–16, respectively, in arXiv:2406.12657v1. The forward-only waiting-time expression is

$$\mathcal{E}_{\mathcal{T}} = \sum_{I,J} n_I^{\mathcal{T}} \int_0^\infty \Psi_{I\to J}^{\mathcal{T}}(t) \log \frac{\Psi_{I\to J}^{\mathcal{T}}(t)}{\Psi_{J\to I}^{\mathcal{T}}(t)}\, dt,$$

where $n_I^{\mathcal{T}}$ is the phase-averaged frequency of transition I. Here $\Psi_{I\to J}^{\mathcal{T}}$ is the conditional density of the next observed event and its waiting time, averaged over the starting phase with event weights $p_i(\theta)Q_{ij}(\theta)/(\mathcal{T}n_I^{\mathcal{T}})$, not with uniform phase weights. The next proposition concerns this expression's limit, not the conjectured finite-period inequality.

Proposition 5.1 (Locality under uniform slow driving)**.** *Let $Q_{\mathcal{T}}(t) = q(t/\mathcal{T})$, where q is a continuously differentiable, period-one generator on a finite fixed connected undirected graph. Assume both directional rates on every edge lie in a fixed interval $[a, b]$, $0 < a \le b < \infty$, and every $q(s)$ is reversible with stationary law $\pi(s)$. Observe a fixed nonempty set of directed edges closed under reversal, with no detector memory or missed events on those edges. Then, as $\mathcal{T} \to \infty$,*

$$\mathcal{E}_{\mathcal{T}} \longrightarrow 0, \qquad \bar{\sigma}_{\mathcal{T}} \longrightarrow 0.$$

For each fixed waiting time $t > 0$, the intensity-weighted pair densities $h^{\mathcal{T}}_{IJ}(t) = n^{\mathcal{T}}_I \Psi^{\mathcal{T}}_{I\to J}(t)$ converge to densities $h^{\infty}_{IJ}(t) = h^{\infty}_{\widetilde{J}\widetilde{I}}(t)$.

Proof. The fixed graph and uniform rate bounds give a time-uniform minorization over an interval of fixed length: prescribe a path of at most $|S| - 1$ edges between any two states, place its jumps in successive subintervals, and use the lower rate and upper escape-rate bounds. Consequently the transition operators contract zero-mass signed measures exponentially, with constants independent of $\mathcal{T}$. The stationary law $\pi(s)$ is continuously differentiable and has a positive uniform lower bound. Variation of constants for $p_{\mathcal{T}}(t) - \pi(t/\mathcal{T})$, whose forcing $-\pi'(t/\mathcal{T})/\mathcal{T}$ has zero total mass because $\sum_i \pi_i(s) = 1$, gives

$$\sup_t \|p_{\mathcal{T}}(t) - \pi(t/\mathcal{T})\| = O(\mathcal{T}^{-1}).$$

Thus the flux differences and their log-ratios in the entropy-production formula are both $O(\mathcal{T}^{-1})$, uniformly, and $\bar{\sigma}_{\mathcal{T}} = O(\mathcal{T}^{-2})$.

Let K_s have the diagonal of $q(s)$, its hidden off-diagonal transitions, and zero off-diagonal entries for observed transitions. Its semigroup describes evolution killed at the next observed event. Writing $U_{\mathcal{T}}(\theta, \theta + t)$ for the corresponding time-dependent killed propagator, the exact finite-period density for $I = (i, j)$, $J = (k, l)$ is

$$h^{\mathcal{T}}_{IJ}(t) = \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} p_i(\theta) Q_{ij}(\theta) (U_{\mathcal{T}}(\theta, \theta + t))_{jk} Q_{kl}(\theta + t)\, d\theta.$$

Continuity of finite-dimensional propagators on every fixed time interval, uniformly in $s = \theta/\mathcal{T}$, gives

$$h^{\infty}_{IJ}(t) = \int_0^1 \pi_i(s) q_{ij}(s) (e^{tK_s})_{jk} q_{kl}(s)\, ds.$$

Detailed balance, including $\pi_j(s)(e^{tK_s})_{jk} = \pi_k(s)(e^{tK_s})_{kj}$, proves the reversal identity for this limit.

It remains to justify integration over arbitrarily long waiting times. From every state there is a uniformly positive probability of an observed event during an interval of fixed length, by the same path argument. The killed propagators therefore have a uniform exponential tail, giving $h^{\mathcal{T}}_{IJ}(t) \le Ce^{-\kappa t}$. If a pair is reachable through hidden edges, let m be the shortest hidden-path length between j and k. Prescribing just that path gives $h^{\mathcal{T}}_{IJ}(t) \ge ct^m e^{-\Lambda t}$; the reversed pair has the same support and path length. These bounds hold for all sufficiently large $\mathcal{T}$, with constants depending only on the fixed model. Unsupported pairs are identically zero in both directions. On supported pairs they imply

$$\left| h^{\mathcal{T}}_{IJ}(t) \log \frac{h^{\mathcal{T}}_{IJ}(t)}{h^{\mathcal{T}}_{\widetilde{J}\widetilde{I}}(t)} \right| \le C' e^{-\kappa t} (1 + t + |\log t|),$$

an integrable bound. Dominated convergence sends the sum of these intensity-pair relative entropies to zero.

Finally, consecutive-event counts have equal incoming and outgoing marginals: $\sum_J \int_0^\infty h^{\mathcal{T}}_{IJ}(t)\, dt = n^{\mathcal{T}}_I$ and $\sum_I \int_0^\infty h^{\mathcal{T}}_{IJ}(t)\, dt = n^{\mathcal{T}}_J$. For the incoming identity, every observed event has an almost-surely unique observed predecessor by the waiting-time tail bound. Decompose $p_k(\theta')$ according to the last observed event before θ', multiply by $Q_{kl}(\theta')$, and average θ' over a period. The estimator is exactly

$$\mathcal{E}_{\mathcal{T}} = \sum_{I,J} \int_0^\infty h^{\mathcal{T}}_{IJ} \log \frac{h^{\mathcal{T}}_{IJ}}{h^{\mathcal{T}}_{\widetilde{J}\widetilde{I}}}\, dt - \sum_I n^{\mathcal{T}}_I \log \frac{n^{\mathcal{T}}_I}{n^{\mathcal{T}}_{\widetilde{I}}}.$$

Here $n^{\mathcal{T}}_I \to \int_0^1 \pi_i(s) q_{ij}(s)\, ds$, and these limits agree for I and $\widetilde{I}$. The second sum also tends to zero. □

The uniform hypotheses make the fixed-waiting-time limit and its integration explicit. They do not cover rates that vanish, a graph that changes with the period, or observation windows that grow proportionally to the period. The last case is precisely the mechanism of Theorem 4.1; it does not refute Maier et al.'s distinct finite-period conjecture. For the two-state witness itself, the only supported consecutive pairs are $(0\to1, 1\to0)$ and $(1\to0, 0\to1)$. Both are self-reverse under $(I, J) \mapsto (\widetilde{J}, \widetilde{I})$, so $\mathcal{E}_{\mathcal{T}} = 0$ at every period. The proposition also covers graphs with hidden edges, where observed pairs need not be self-reverse.

6 Summary of information requirements

conclusion	forward-only, phase erased	additional information
a lower bound on $\bar{\sigma}$	yes, from the limiting arrow rate if the protocol is time-symmetric and O is covariant (T1)	otherwise, for a universal quantitative bound from this arrow alone (T2); qualitative, model-specific implications survive (Remark 4.4)
a finite-window bound	yes, minus the allowance $\bar{B}$ (model-dependent; bounded if $p \geq p_{\min} > 0$)	–
a general periodic protocol	no universal quantitative bound from this arrow alone (T2)	Harunari–Fiore–Barato's $\sigma \geq \hat{\sigma} - X$ uses reversed-protocol data; Maier et al.'s Eq. 19 uses phase-resolved occupations and rates
a forward-only pairwise estimator	Maier et al.'s Eq. 18, conjectured; consistent with Proposition 5.1	–

An observation with memory, event selection or dead time requires its own reversal-covariance check. The time-symmetric bound does not automatically apply to an arbitrary detector model.

7 Computational checks and limitations

The supplied numerical route uses Python 3.11 and mpmath 1.3.0 at 30 digits with deterministic quadrature. For the piecewise-constant drive $u = 1/4, 3/4, 1/4$ on three successive intervals of length $h = \log 2$, it reproduces the periodic switching-time populations $(11/28, 9/28, 15/28)$, the mean $3/14$ of $X(2h) - X(h)$, and the entropy production $3\log 3/7$ per cycle. The six-period asymmetric study is in Table 1. A symmetric control with $(m, e_1, e_2, \varphi) = (0.45, 0.25, 0.15, \pi)$ checks $\mathrm{KL}_3 \leq L\bar{\sigma} + \bar{B}$ at periods $10, 40, 160, 640$. It checks a consequence of the full-path bound, not the four-sample arrow directly. At period 10, its comparator is approximately $9.0 \cdot 10^{-7}$, against $L\bar{\sigma} = 0.418$ and $\bar{B} = 0.083$.

An earlier symmetric control used $e_2 = 0.2$, making its rate zero at phase π, outside the declared positive-rate domain. Numerical evaluation returned infinity and its acceptance test did not reject a non-finite value. That block is invalid; its original result is retained and identified in the supplement's history. The replacement control requires finite outputs and a positive rate range. The asymmetric values are unchanged. The numerical infinity is not a divergence theorem: the old drive obeys $u(\pi + z) = 21z^2/40 + O(z^4)$, so for finite period its entropy integrand has only an integrable logarithmic singularity there. This does not put the old run inside the positive-rate hypothesis.

Exact probability checks. The four-sample table in Appendix A is reconstructed by a Wolfram Engine 14.3 route and a separately coded rational Laurent-polynomial route over $\mathbb{Q}(\sqrt{-3})$, with logarithms evaluated at 50 digits on the latter. Positivity, normalization, both marginal identities and the strict relative-entropy inequality are checked. The two routes agree on c_3^* and c_4^* to 45 digits. A first corroborating grid check used binary floating-point coefficients and failed at the exact upper endpoint $9/10$; its failed record and the exact-coefficient replacement are both supplied. These high-precision numbers are not interval enclosures. The continuous bounds and the fixed-arrow existence statement follow from the written proofs, not from a sampled grid.

The supplement [6] identifies commands, dependencies, result files and the interpretation of earlier failures. The locality proposition is analytic, not an output of these numerical checks. All conclusions concern the stated Markov and observation models; no physical measurement, detector calibration or priority claim is made.

Authorship, assistance and rights

Luqman Mahmood directed this work and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted with mathematical development, computation, checking and editing. Internal and automated checks are not external peer review. Computational tests do not replace proofs of universal statements, and no physical identification or experiment is implied. The paper and mathematical data are licensed CC BY 4.0. Software source and executable mathematical verification scripts are excluded from that grant; their rights are reserved unless explicitly stated otherwise.

References

[1] G. E. Crooks, Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences, *Phys. Rev. E* **60** (1999), 2721–2726. https://doi.org/10.1103/PhysRevE.60.2721.

[2] R. Kawai, J. M. R. Parrondo and C. Van den Broeck, Dissipation: The phase-space perspective, *Phys. Rev. Lett.* **98** (2007), 080602. https://doi.org/10.1103/PhysRevLett.98.080602.

[3] A. Gomez-Marin, J. M. R. Parrondo and C. Van den Broeck, Lower bounds on dissipation upon coarse graining, *Phys. Rev. E* **78** (2008), 011107. https://doi.org/10.1103/PhysRevE.78.011107.

[4] A. M. Maier, J. Degünther, J. van der Meer and U. Seifert, Inferring kinetics and entropy production from observable transitions in partially accessible, periodically driven Markov networks, *J. Stat. Phys.* **191** (2024), 104. https://doi.org/10.1007/s10955-024-03315-7; preprint https://arxiv.org/abs/2406.12657.

[5] P. E. Harunari, C. E. Fiore and A. C. Barato, Inference of entropy production for periodically driven systems, *Phys. Rev. E* **110** (2024), 064126. https://doi.org/10.1103/PhysRevE.110.064126; preprint https://arxiv.org/abs/2406.12792.

[6] L. Mahmood, *Phase-erased entropy production: computational supplement*, 2026, distributed with this paper; see `supplement/README.md` and `supplement/HISTORY.md`. Source and result paths cited above are relative to that supplement. DOI: https://doi.org/10.5281/zenodo.23049617.

A Exact quasi-static laws at the witness

Copied from `results/EP1B_ADDENDUM_1_CHECK_R2.json`; the Wolfram route prints the same rationals. Drive $(m, e_1, e_2, \varphi) = (9/20, 1/4, 1/5, \pi/3)$, phases $(0, \pi/3, 2\pi/3, \pi)$. For each record x the table gives the forward cell $P_4(x)$ and the reversed cell $R_4(x) = P_4(\text{reverse of } x)$.

x	P4(x)	R4(x)	x	P4(x)	R4(x)
0000	50681/640000	50681/640000	0001	51739/640000	51739/640000
0010	42799/640000	45799/640000	0011	51981/640000	48981/640000
0100	45799/640000	42799/640000	0101	28981/640000	31981/640000
0110	47121/640000	47121/640000	0111	32899/640000	32899/640000
1000	51739/640000	51739/640000	1001	43041/640000	43041/640000
1010	31981/640000	28981/640000	1011	28039/640000	31039/640000
1100	48981/640000	51981/640000	1101	31039/640000	28039/640000
1110	32899/640000	32899/640000	1111	20281/640000	20281/640000

Triple comparator, coordinates $(0, \pi/3, \pi)$: P3 = (2337, 2593, 2323, 1547, 2093, 1777, 2047, 1283)/16000 and

R3 = (2412, 2518, 2248, 1622, 2018, 1852, 2122, 1208)/16000 on 000, 001, ..., 111.

c3 = 0.000738887716627336479919968717667278764901883127

c4 = 0.001534419994182709441167521655042144619538994l2

c4 - c3 = 0.000795532277555372961247552937375